

# Assignment No 06

Name: Om Varma

Roll No: 28

PRN No: 1252010028

## Question:

Publication, Book, and Magazine Using composition

Write a program to:

1. objects of the classes.
1. and display the number of copies ordered for each book.
2. and display the total sales of all publications.

## Code:

```
#include<iostream>
#include<iomanip>
#include<limits>
#include<cctype>
using namespace std;

class Publication{
public:
    string title;
    double price;
    int copies;

    void getData(){
        while(true){
            cout<<"Enter Title: ";
            getline(cin,title);

            bool valid = !title.empty();
            for(char c : title){
                if(!isalpha(c) && c!=' '){
                    valid=false;
                    break;
                }
            }

            if(valid) break;
            cout<<"Invalid Title! Letters only.\n";
        }

        while(true){
            cout<<"Enter Price: ";
            cin>>price;

            if(!cin.fail() && price>0)
                break;
        }
    }
};
```

```

        cout<<"Invalid Price!\n";
        cin.clear();
        cin.ignore(numeric_limits<streamsize>::max(),'\n');
    }

    while(true){
        cout<<"Enter Copies Sold: ";
        cin>>copies;

        if(!cin.fail() && copies>=0)
            break;

        cout<<"Invalid Copies!\n";
        cin.clear();
        cin.ignore(numeric_limits<streamsize>::max(),'\n');
    }
}

double saleCopy(){
    return price * copies;
}

};

class Book{
public:
    Publication p;
    string author;
    int orderedCopies;

    void getData(){
        p.getData();
        cin.ignore(numeric_limits<streamsize>::max(), '\n');

        while(true){
            cout<<"Enter Author: ";
            getline(cin,author);

            bool valid = !author.empty();

            for(char c:author){
                if(!isalpha(c) && c!=' '){
                    valid=false;
                    break;
                }
            }

            if(valid) break;

            cout<<"Invalid Author! Letters only.\n";
        }
    }

    void orderCopies(){
        while(true){

```

```

        cout<<"Enter Copies Ordered: ";
        cin>>orderedCopies;

        if(!cin.fail() && orderedCopies>=0)
            break;

        cout<<"Invalid Copies!\n";
        cin.clear();
        cin.ignore(numeric_limits<streamsize>::max(), '\n');
    }
}

void display(){
    cout<<"\n";
    cout<<"----- BOOK DETAILS -----\n";

    cout<<left<<setw(20)<<"Title:"<<p.title<<endl;
    cout<<left<<setw(20)<<"Author:"<<author<<endl;
    cout<<left<<setw(20)<<"Price:"<<p.price<<endl;
    cout<<left<<setw(20)<<"Copies Sold:"<<p.copies<<endl;
    cout<<left<<setw(20)<<"Copies Ordered:"<<orderedCopies<<endl;
    cout<<left<<setw(20)<<"Sales:"<<p.saleCopy()<<endl;
}

int getOrdered(){
    return orderedCopies;
}

double getSales(){
    return p.saleCopy();
}
};

class Magazine{
public:
    Publication p;
    int orderQty;
    int issue;

    void getData(){
        p.getData();
    }

    void orderQtyFunc(){
        while(true){
            cout<<"Enter Order Quantity: ";
            cin>>orderQty;

            if(!cin.fail() && orderQty>=0)
                break;

            cout<<"Invalid Quantity!\n";
            cin.clear();
            cin.ignore(numeric_limits<streamsize>::max(), '\n');

```

```

    }
}

void currentIssue(){
    while(true){
        cout<<"Enter Current Issue: ";
        cin>>issue;

        if(!cin.fail() && issue>0)
            break;

        cout<<"Invalid Issue!\n";
        cin.clear();
        cin.ignore(numeric_limits<streamsize>::max(), '\n');
    }
}

void receiveIssue(){
    cout<<"Issue Received\n";
}

void display(){
    cout<<"\n";
    cout<<"----- MAGAZINE DETAILS ----- \n";

    cout<<left<<setw(20)<<"Title:"<<p.title<<endl;
    cout<<left<<setw(20)<<"Price:"<<p.price<<endl;
    cout<<left<<setw(20)<<"Copies Sold:"<<p.copies<<endl;
    cout<<left<<setw(20)<<"Order Quantity:"<<orderQty<<endl;
    cout<<left<<setw(20)<<"Issue:"<<issue<<endl;
    cout<<left<<setw(20)<<"Sales:"<<p.saleCopy()<<endl;
}

double getSales(){
    return p.saleCopy();
}

};

int main(){
    Book b;
    Magazine m;

    cout<<"Enter Book Details\n";
    b.getData();
    b.orderCopies();

    cin.ignore(numeric_limits<streamsize>::max(), '\n');
    cout<<"\nEnter Magazine Details\n";
    m.getData();
    m.orderQtyFunc();
    m.currentIssue();
    m.receiveIssue();

    b.display();
}

```

```

m.display();

double totalSales = b.getSales() + m.getSales();

cout<<"\n-----\n";
cout<<left<<setw(20)<<"Copies Ordered:"<<b.getOrdered()<<endl;
cout<<left<<setw(20)<<"Total Sales:"<<totalSales<<endl;
cout<<"-----\n";

return 0;
}

```

Output:

```

PS D:\STUDY\B.Tech\SEM IV\OOP\Assignment 6> .\a.exe
Enter Book Details
Enter Title: words of wisdom
Enter Price: 5000
Enter Copies Sold: 99
Enter Author: Om Varma
Enter Copies Ordered: 70

Enter Magazine Details
Enter Title: Words of Wisdom
Enter Price: 90
Enter Copies Sold: 0
Enter Order Quantity: 0
Enter Current Issue: 10
Issue Received

----- BOOK DETAILS -----
Title:           words of wisdom
Author:          Om Varma
Price:           5000
Copies Sold:     99
Copies Ordered:  70
Sales:           495000

----- MAGAZINE DETAILS -----
Title:           Words of Wisdom
Price:           90
Copies Sold:     0
Order Quantity:  0
Issue:           10
Sales:           0

-----
Copies Ordered:  70
Total Sales:     495000

```